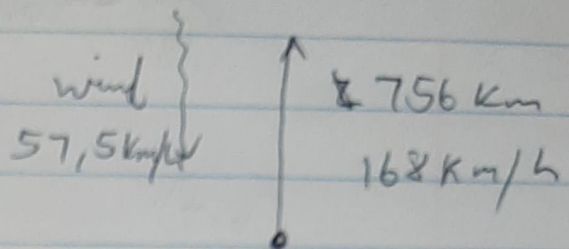


Semester Test FSK 116

PART A

Q1

1. a) $168 - 57,5$
 $= 110,5 \text{ km/h}$ relative to ground

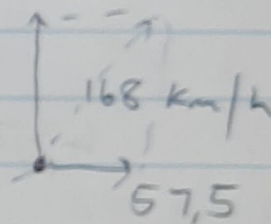


$\frac{756}{110,5} = 6,84 \text{ h}$

b) $168 + 57,5$
 $= 225,5 \text{ km/h}$ relative to ground

$\frac{756}{225,5} = 3,35 \text{ h}$

c) $U_{R0}^2 = 168^2 + 57,5^2$
 $U_{R0} = 177,57 \text{ km/h}$



$\frac{756}{177,57} = 4,26 \text{ h}$

Q2

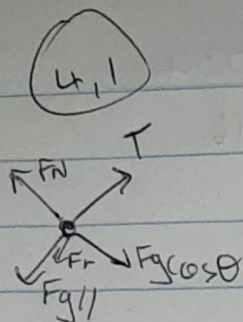
$\Delta x = \left(\frac{U_f + U_i}{2} \right) \Delta t$
 $0,226 = \frac{250}{2} \Delta t$
 $\Delta t = 1,808 \times 10^{-3} \text{ s}$

$U_f = 0$
 $U_i = 250 \text{ m/s}$
 $m = 0,0115 \text{ kg}$
 $\Delta x = 0,226 \text{ m}$

$F_{ext} \Delta t = m U_f - m U_i$
 $F_{ext} = \frac{0 - 2,875}{1,808 \times 10^{-3}}$

$= 1590,15$
 $= 1590$

Q3



$$F_{net} = m \cdot a$$

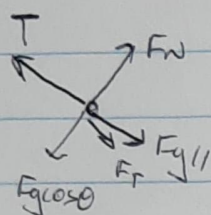
$$T - F_{g||} - F_r = (4,1) \cdot (0,83)$$

$$T = (3,403) + (23,046) + (32,91 \text{ Ms})$$

--- (1)

$$F_N = mg \cos \theta = 53,786$$

6,7



$$F_{net} = m \cdot a$$

$$T = (5,561) + (37,661) + (53,786 \text{ Ms})$$

--- (2)

~~43,222 + 53,786 Ms~~

$$43,222 + 53,786 \text{ Ms} = 26,449 + 32,91 \text{ Ms}$$

$$16,773 = 32,91 \text{ Ms} - 53,786 \text{ Ms}$$

$$16,773 = -20,876 \text{ Ms}$$

$$\text{Ms} = 0,80$$

$$F_N = F_g \cos \theta = 53,786$$

$$F_{net} = m \cdot a$$

$$(F_r + F_{g||}) - T = 4,79 \cdot 5,561$$

$$-T = 5,561 - F_r - F_{g||}$$

$$T = -5,561 + F_r + F_{g||}$$

$$= -5,561 + (53,786 \text{ Ms}) + 37,661$$

$$T = 32,1 + (53,786 \text{ Ms}) \quad \dots (2)$$

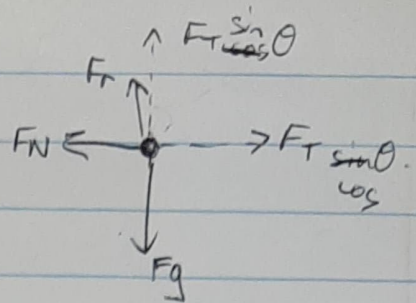
$$26,449 + (32,91 \text{ Ms}) = 32,1 + (53,786 \text{ Ms})$$

$$-20,876 \text{ Ms} = 5,651$$

$$\text{Ms} = 0,27$$

$$T = 32,1 + (53,786 \times 0,27) = 46,6 \text{ N}$$

Q4 $F_{net} = m \cdot a$



$$F_r + F_T \sin \theta = F_g$$

$$0,18 P + 0,766 P = 23,52$$

$$0,946 P = 23,52$$

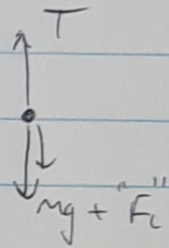
$$P = 24,86 \text{ N}$$

Q5

$$F_c = \frac{m v^2}{R}$$

$$1000 = \frac{(87,5)(v)^2}{10,5}$$

$$v = 10,95$$



$$857,5 + 450,187$$

$$= 1307,6875 \text{ N}$$

Q6